

GENOMIC PIPELINE REPORT

Applying a Genomics Pipeline to Determine the Role of Increased blaOXA-48 Copy Numbers and OmpK36 Loss in Carbapenem Resistance in *Klebsiella pneumoniae*

Based on data from: Meekes et al., Antimicrobial Agents and Chemotherapy, 2025

Abstract

Antibiotic resistance in *Klebsiella pneumoniae* constitutes a major public health threat driven by complex genetic and regulatory mechanisms. The objective of this study was to analyse hybrid genome sequencing data generated by Illumina short-read and Nanopore long-read technologies in order to elucidate the antimicrobial resistance (AMR) mechanisms of clinical and laboratory-derived *K. pneumoniae* strains selected under meropenem pressure. The analytical workflow comprised hybrid *de novo* genome assembly, antimicrobial resistance gene (ARG) annotation, quantification of plasmid copy numbers (PCN), and prediction of structural variations, including regulatory point mutations and insertion sequence (IS) events. Computational analysis demonstrated that high-level carbapenem resistance in the isolates is governed by the combined effects of increased PCN of plasmids carrying *bla*OXA-48 or *bla*CTX-M-15 genes and inactivation of the major porin OmpK36 through IS1 transposition or large genomic deletions. In addition, the study identified regulatory mutations responsible for elevated PCN, notably single-nucleotide polymorphisms within the *copA* antisense RNA of IncFII plasmids and structural rearrangements in IncR plasmids. Collectively, these findings indicate that a carefully designed hybrid genomics pipeline can accurately reconstruct the principal molecular mechanisms of carbapenem resistance, although complementary proteomic data remain essential for the complete validation of expression-level phenotypes.

1. Introduction

The global dissemination of carbapenem-resistant *Klebsiella pneumoniae* has emerged as one of the most pressing challenges in clinical microbiology and infectious-disease management. Carbapenems remain among the last-resort agents for treating serious infections caused by multidrug-resistant Enterobacterales; consequently, any reduction in their efficacy severely limits therapeutic options and is associated with increased morbidity and mortality. Resistance to carbapenems arises principally through the acquisition of carbapenemase genes, of which *Klebsiella pneumoniae* carbapenemase (KPC), New Delhi metallo- β -lactamase (NDM) and OXA-48-like enzymes are the most frequently encountered in clinical settings.

Enzymes belonging to the OXA-48 family exhibit only weak carbapenem-hydrolysing activity. As a result, isolates that harbour *bla*OXA-48 typically display only modest elevations in carbapenem minimum inhibitory concentrations (MICs). High-level resistance, however, can develop when *bla*OXA-48 is combined with additional mechanisms, most notably the loss or severe reduction of outer-membrane porins that normally facilitate antibiotic entry into the

bacterial cell. Parallel pathways of resistance exist in non-carbapenemase-producing carbapenem-resistant Enterobacterales (non-CP-CRE). In these organisms, the synergistic combination of porin deficiency and elevated expression of extended-spectrum β -lactamases (ESBLs) or AmpC cephalosporinases is sufficient to push carbapenem MICs above clinical breakpoints.

Surveillance programmes have shown that non-CP-CRE account for a substantial fraction (19–52 %) of carbapenem-non-susceptible Enterobacterales recovered from clinical specimens. Because conventional PCR-based assays interrogate only a limited panel of known resistance genes, they are inherently incomplete for characterising the full spectrum of genetic events that underlie phenotypic resistance. Whole-genome sequencing (WGS), particularly when performed with hybrid short- and long-read technologies, offers a far more comprehensive view: it simultaneously captures acquired resistance genes, chromosomal mutations, structural variations and quantitative changes in plasmid dosage.

The present study therefore sought to apply a custom hybrid genomics pipeline to a well-characterised collection of clinical and laboratory-adapted *K. pneumoniae* isolates originally described by Meekes et al. The dual aims were (i) to reconstruct the resistome and plasmid content of the clinical isolates, and (ii) to identify the precise genetic events—point mutations, insertion-sequence transposition and plasmid-copy-number variation—that drive the rapid emergence of high-level carbapenem resistance under selective pressure. By comparing the in-silico predictions with the published proteogenomic data, the study also evaluates the extent to which a genomics-only workflow can recapitulate findings that originally required both DNA- and protein-level analyses.

2. Materials and Methods

2.1. Isolate Collection and Sequencing

The primary dataset comprised fifteen clinical *K. pneumoniae* isolates recovered during an outbreak at the Regional Institute of Gastroenterology and Hepatology in Cluj-Napoca, Romania, together with four laboratory-derived mutants obtained after sequential meropenem exposure. The clinical isolates had been selected to represent the spectrum of carbapenem MICs and multi-locus variable-number tandem-repeat analysis (MLVA) types present in the larger outbreak collection of 74 isolates. Total genomic DNA was extracted with MagNA Pure 96 DNA and Viral NA Small Volume kits for Illumina sequencing and with Qiagen Genomic-tip 20/G columns for Nanopore sequencing. Illumina libraries generated 150-nucleotide paired-end reads, providing high per-base accuracy. Nanopore libraries were prepared with the Rapid Barcoding kit and sequenced on a GridION instrument equipped with R9.4 flow cells, yielding long reads capable of spanning repetitive regions and resolving complete circular plasmids.

2.2. Bioinformatics Pipeline Overview

A highly scalable, custom bioinformatics pipeline was developed and executed entirely within a Linux (Ubuntu) command-line environment. The workflow integrates seven sequential modules that progress from raw-read quality control through hybrid assembly, functional annotation, quantitative plasmid analysis, structural-variant detection and phylogenomic reconstruction (Figure 1). Each module was selected for robustness, reproducibility and compatibility with both short- and long-read data streams.

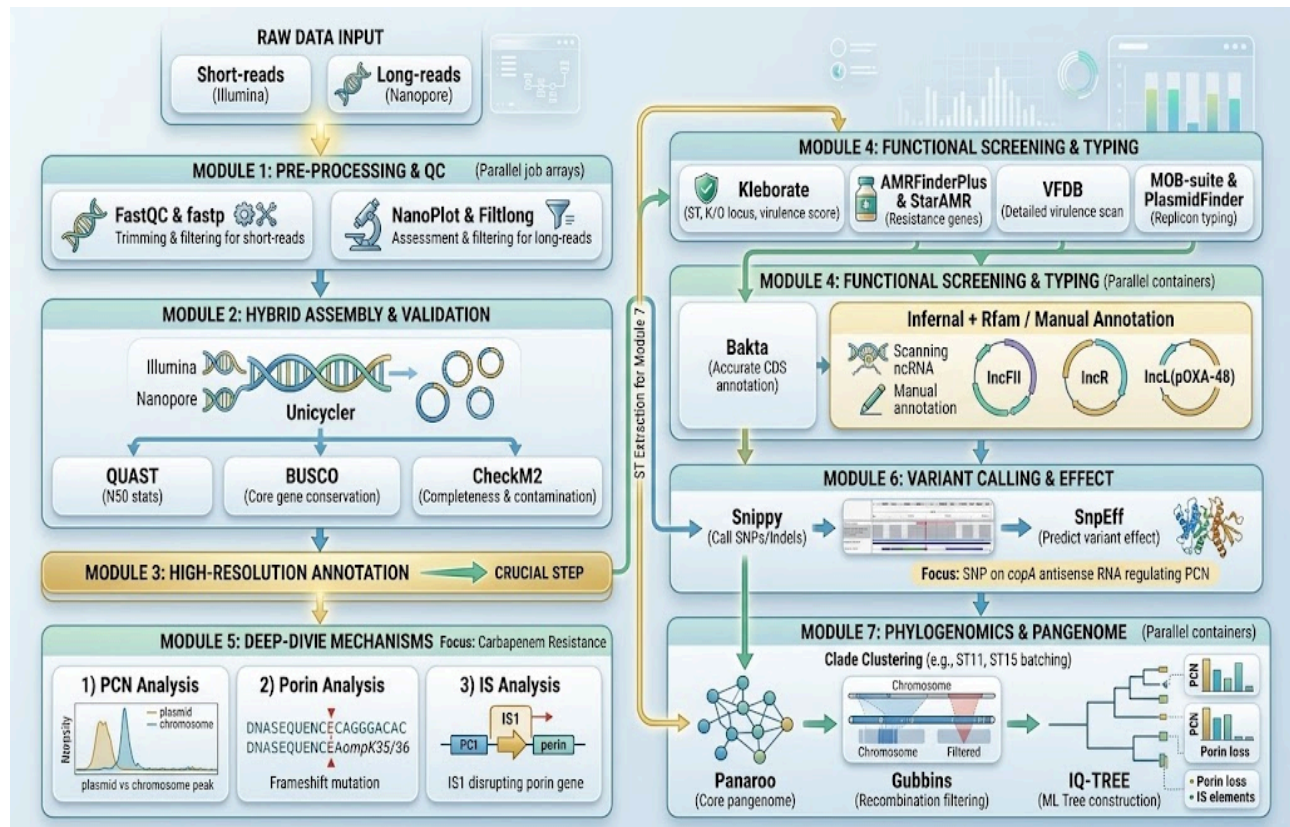


Figure 1. Overview of the custom genomics pipeline. The workflow comprises seven integrated modules ranging from raw hybrid-read pre-processing to advanced phylogenomics and deep-dive molecular mechanism analysis.

2.3. Read Processing and Hybrid Assembly

Short reads were quality-filtered and adapter-trimmed with FastQC and fastp. Long reads were assessed with NanoPlot and filtered with Filtlong to retain the highest-quality subset. Hybrid *de novo* assembly was performed with Unicycler, which leverages the accuracy of Illumina data and the contiguity of Nanopore data to produce complete, circular chromosomal and plasmid sequences. Assembly quality was evaluated with QUAST (contiguity metrics), BUSCO (single-copy orthologue completeness) and CheckM2 (contamination and completeness estimates). All assemblies satisfied the MIMAG criteria for high-quality draft genomes.

2.4. Annotation and Resistome Characterisation

Protein-coding genes were annotated with Bakta. Non-coding RNAs were identified with Infernal against the Rfam database, with particular attention paid to the *copA–copT* antisense-RNA cassette that regulates replication of IncFII plasmids. Sequence types were assigned with Kleborate according to the Pasteur multi-locus sequence typing scheme. Antimicrobial resistance genes and chromosomal point mutations were detected with AMRFinderPlus and StarAMR. Virulence factors were screened against the VFDB database, while plasmid replicons were typed with MOB-suite and PlasmidFinder.

2.5. Quantitative and Structural Analyses

Plasmid copy numbers were calculated by mapping quality-filtered reads to the assembled genome with BWA-MEM and Samtools, then normalising the mean depth of each plasmid replicon against the mean depth of the chromosome. Insertion-sequence elements that potentially disrupt the porin genes *ompK35* and *ompK36* were detected with ISEScan. Single-nucleotide polymorphisms and small indels were called with FreeBayes or Snippy and functionally annotated with SnpEff. Core-genome alignments were generated with Panaroo, recombination was filtered with Gubbins, and maximum-likelihood phylogenies were inferred with IQ-TREE under the best-fit substitution model selected by ModelFinder.

3. Results

3.1. Genomic Assembly and Isolate Characterisation

Hybrid Illumina–Nanopore sequencing produced highly contiguous assemblies for all nineteen genomes (fifteen clinical isolates and four meropenem-adapted mutants). The mean N50 value reached 5.13 Mb, and the majority of chromosomes and plasmids were recovered as single, circular contigs. BUSCO completeness scores exceeded 98.7 % on average, while CheckM2 completeness approached 100 %. Contamination estimates remained extremely low (0.08–0.90 %), well below the 5 % MIMAG threshold for high-quality draft genomes. In-silico multi-locus sequence typing with Kleborate confirmed the epidemiological diversity of the collection, with ST101 and ST15 emerging as the predominant lineages (Figure 2). These high-quality assemblies provided a robust foundation for all subsequent quantitative and structural analyses.

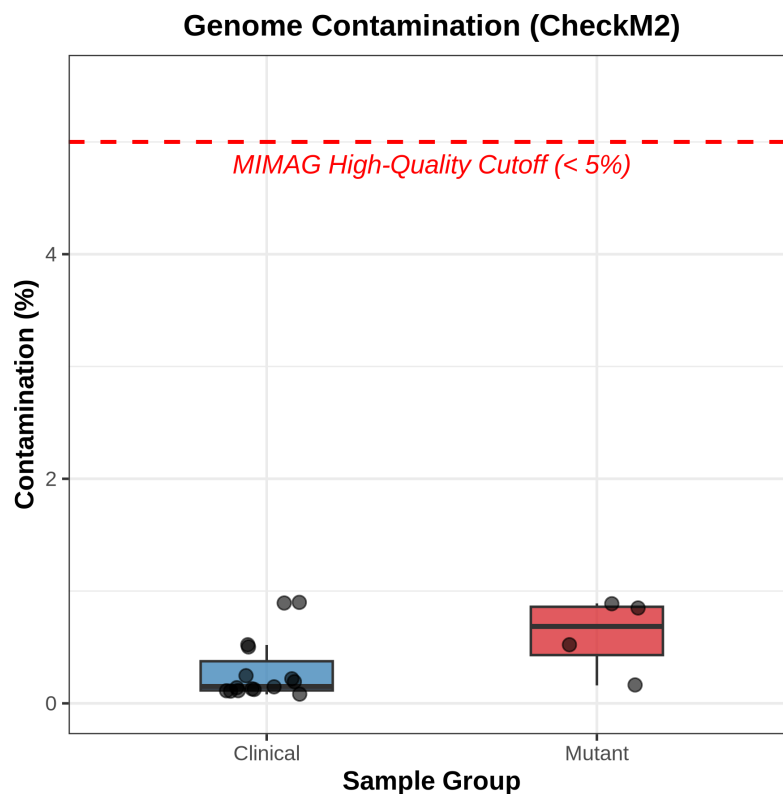


Figure 2C. CheckM2 contamination estimates. The red dashed line indicates the MIMAG 5% threshold.

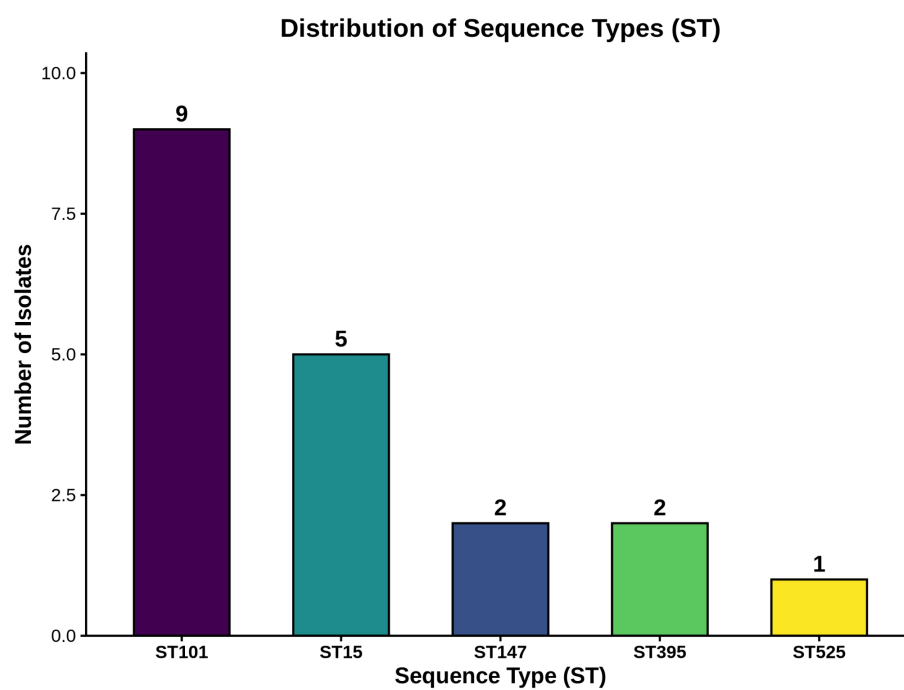


Figure 2D. Distribution of sequence types (STs) determined by Kleborate.

3.2. Comprehensive Resistome Profiling

Functional annotation revealed an extensive and diverse resistome across the entire isolate collection. Every genome carried a multidrug-resistant profile driven by the cumulative acquisition of resistance determinants on both the chromosome and mobile genetic elements.

3.2.1. Beta-Lactamase Determinants

The extended-spectrum β -lactamase gene *bla*CTX-M-15 was detected in the large majority of isolates and frequently occurred in two or three genomic copies, correlating with high-level resistance to third-generation cephalosporins. The carbapenemase gene *bla*OXA-48 was exclusively localised on conserved IncL(pOXA-48)-type plasmids in ten clinical isolates. Isolate Kpn-15 uniquely co-harboured *bla*OXA-48 and the metallo- β -lactamase gene *bla*NDM-1, providing a potent dual-carbapenemase background that explained its extreme MICs against all tested carbapenems. A dense background of additional β -lactamases—diverse alleles of *bla*SHV, *bla*TEM and *bla*OXA—further reinforced the enzymatic barrier (Figure 3).

3.2.2. Additional Resistance Classes

Aminoglycoside resistance was conferred by a rich array of modifying enzymes (*aac*, *aph* and *ant* family members) and, in isolate Kpn-15, by the 16S rRNA methyltransferase gene *rmtC*. Fluoroquinolone non-susceptibility resulted from the universal presence of the chromosomal RND efflux operons *oqxAB*, plasmid-mediated quinolone-resistance genes (*qnrS1*, *aac(6')-Ib-cr*) and characteristic mutations in the quinolone-resistance-determining regions of *gyrA* and *parC*. Supplementary determinants conferred resistance to fosfomycin (*fosA*), co-trimoxazole (*sul* and *dfr* alleles), tetracyclines, chloramphenicol, macrolides and rifampicin. Laboratory-derived mutants consistently carried a higher overall AMR gene burden than their parental clinical isolates, reflecting adaptive expansion under antibiotic selection.

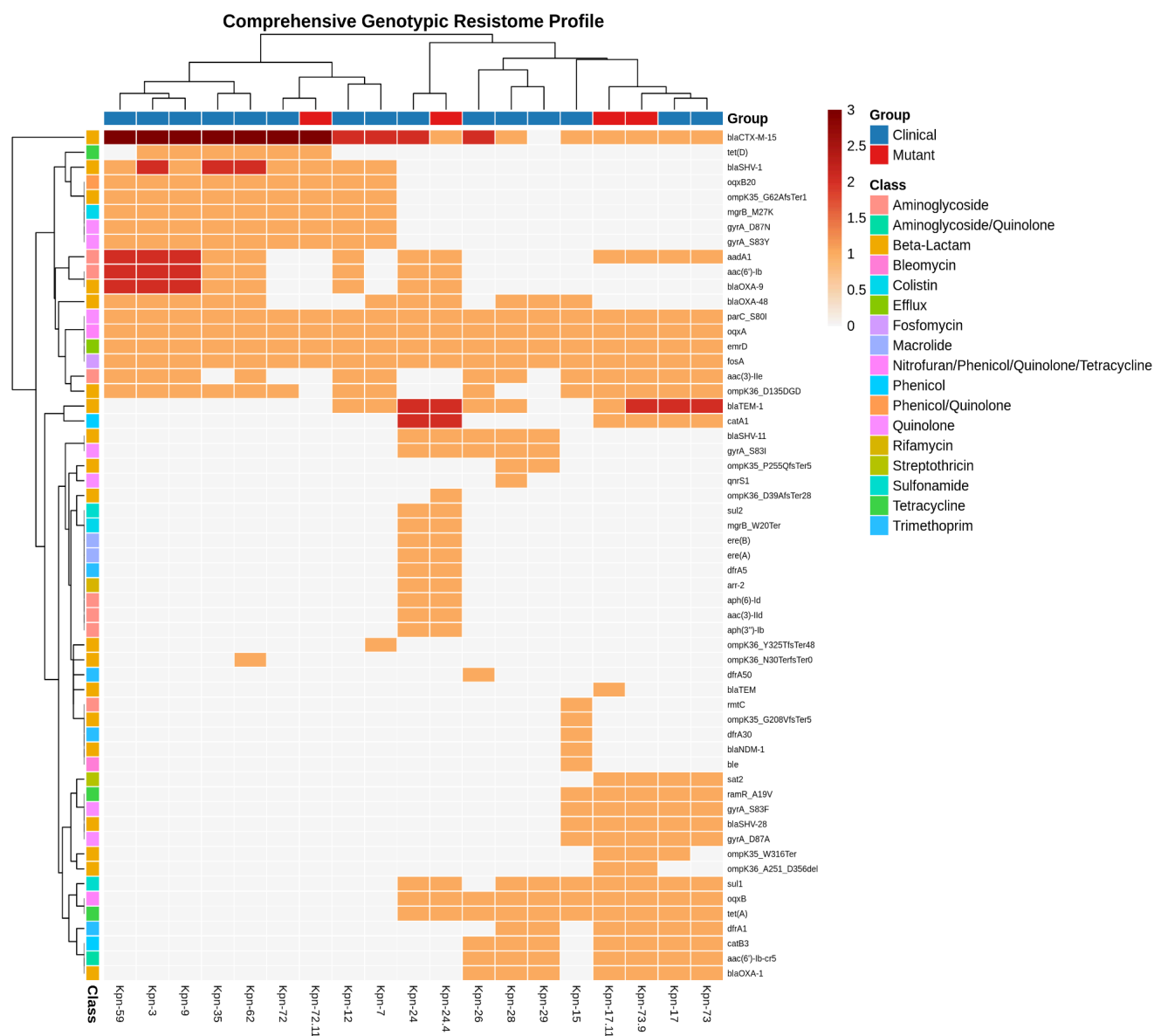


Figure 3A. Heatmap illustrating the distribution and copy-number variation of acquired AMR genes and point mutations across isolates.

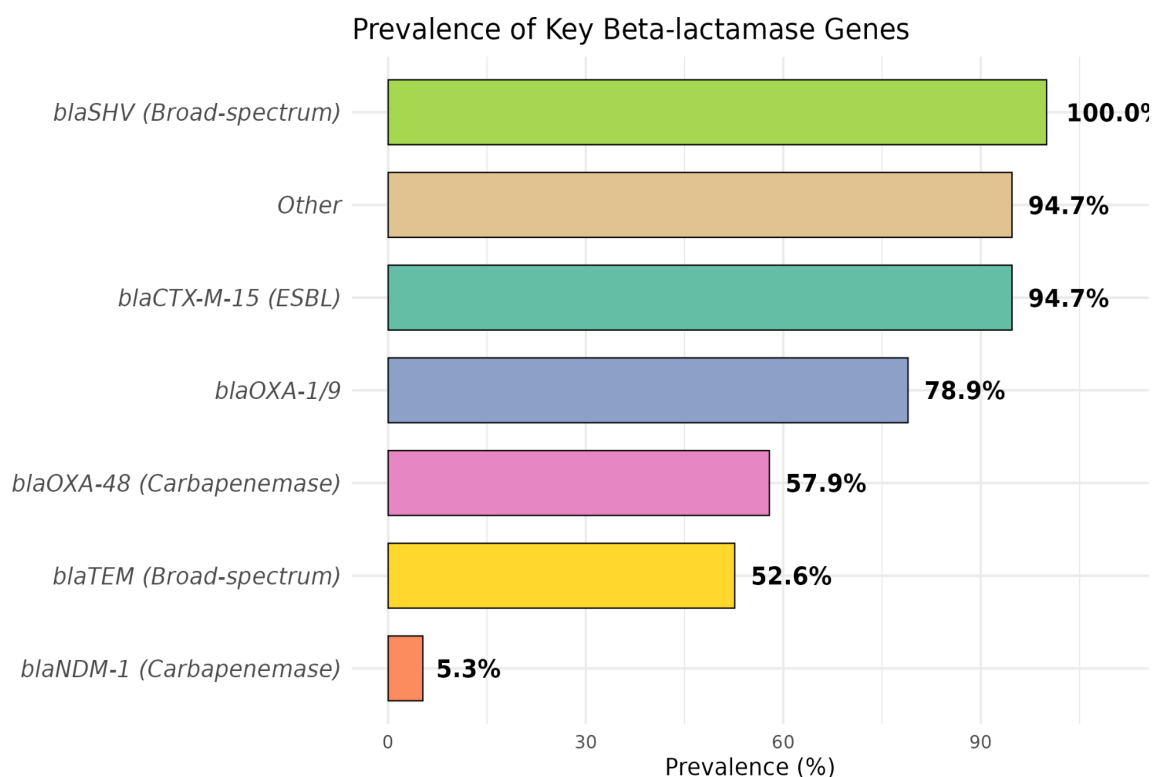


Figure 3B. Prevalence of key beta-lactamase gene families across the isolate collection.

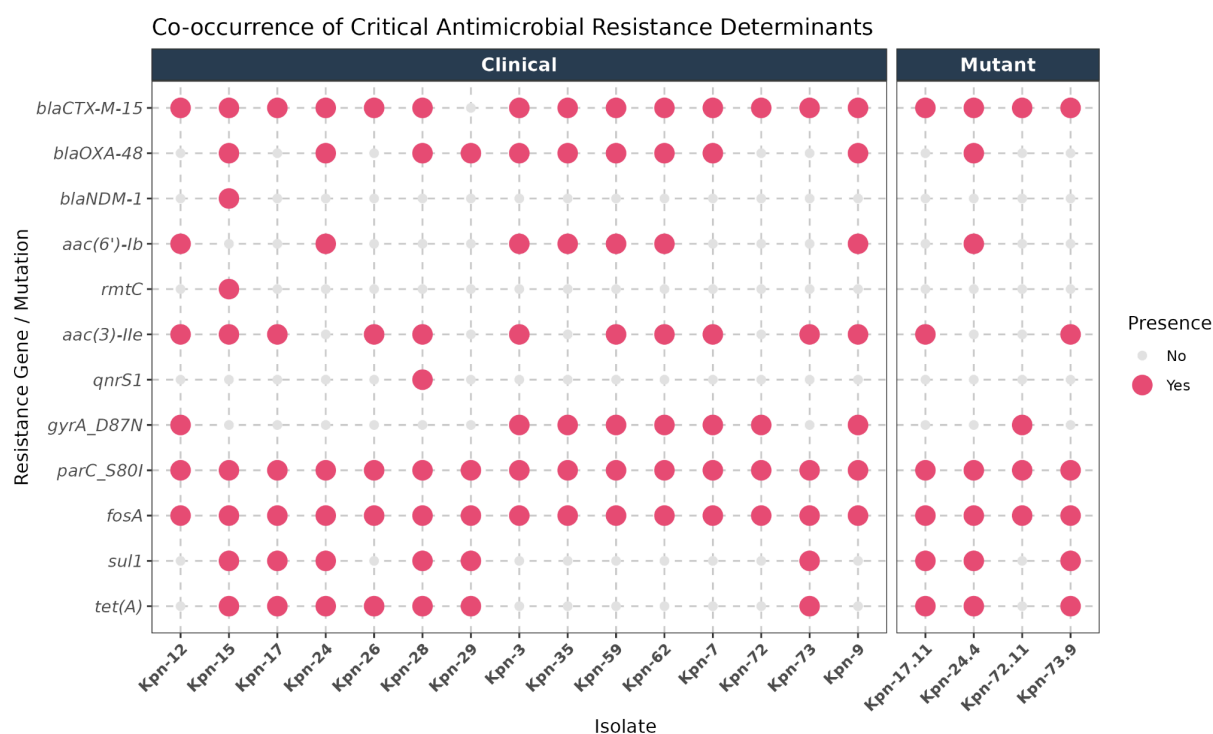


Figure 3C. Co-occurrence of high-impact resistance determinants within individual isolates.

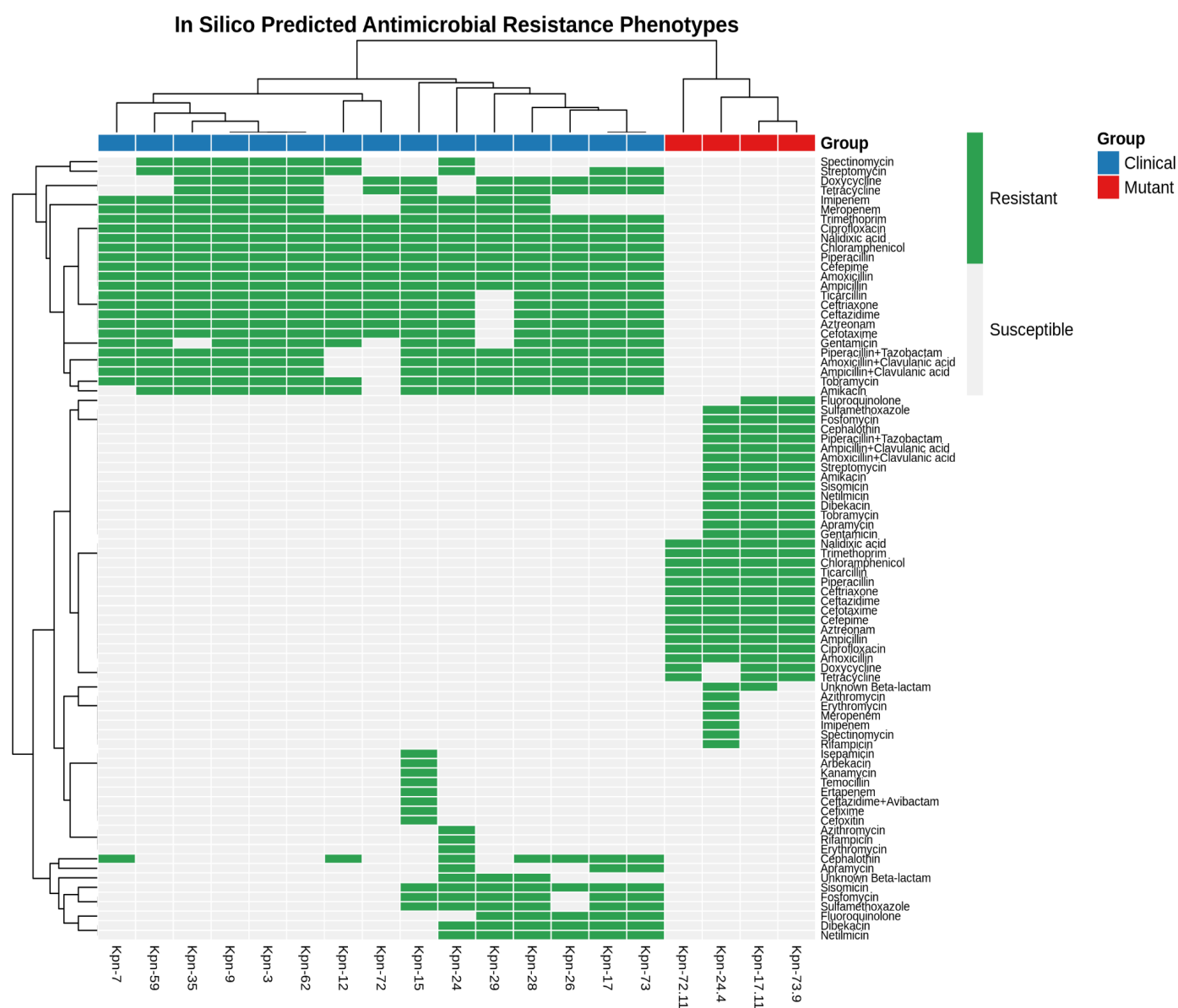


Figure 3D. In silico predicted antimicrobial resistance phenotypes generated by StarAMR.

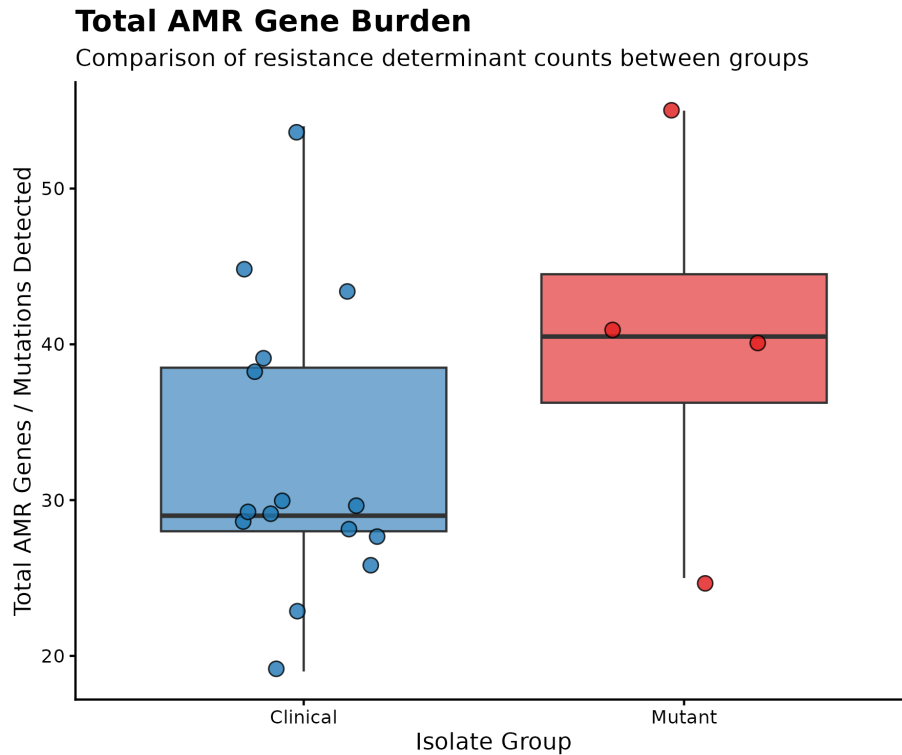


Figure 3E. Comparison of total AMR gene burden between clinical isolates and meropenem-exposed mutants.

3.3. Plasmid Copy Number Variations Across Clinical and Mutant Isolates

Normalisation of plasmid read depth against chromosomal coverage revealed substantial natural variation in plasmid copy number among the clinical isolates. Ten isolates carried *bla*OXA-48 on IncL-type plasmids whose baseline PCN ranged from 0.7 $\times$ to 6.6 $\times$. Remarkably, five of these isolates (Kpn-3, Kpn-24, Kpn-35, Kpn-59 and Kpn-62) shared a 100 % identical IncL(pOXA-48) plasmid sequence, yet their PCNs differed by nearly an order of magnitude, indicating that copy-number control is not determined solely by plasmid sequence. For the non-CP-CRE isolates that relied on *bla*CTX-M-15, baseline PCNs of the cognate IncFII and IncR replicons ranged between 1.2 $\times$ and 2.6 $\times$.

Under sequential meropenem selection the copy numbers of resistance plasmids increased markedly. IncFII plasmids carrying *bla*CTX-M-15 amplified 3.6-fold (mutant 17.11) and 6.6-fold (mutant 73.9). An IncR plasmid in mutant 72.11 increased 5.1-fold, while the IncL plasmid of mutant 24.4 rose 3.2-fold (Figure 4). Absolute sequencing-depth analysis confirmed that chromosomal coverage remained stable (114–304 $\times$) while plasmid depth surged (in one case reaching 1 930 $\times$), verifying that the observed PCN increases represent genuine biological amplification rather than sequencing artefacts.

Table 1. Baseline Copy Numbers of Primary Resistance Plasmids in 15 Clinical Isolates

Clinical Isolate	Primary Resistance Driver	Target Plasmid Replicon	Computed PCN (Relative to Chromosome)
Kpn-15	<i>bla</i> ~OXA-48~ / <i>bla</i> ~NDM-1~	IncL	0.7X
Kpn-35	<i>bla</i> ~OXA-48~	IncL	0.8X
Kpn-59	<i>bla</i> ~OXA-48~	IncL	0.8X
Kpn-24	<i>bla</i> ~OXA-48~	IncL	1.0X
Kpn-9	<i>bla</i> ~OXA-48~	IncL	1.0X
Kpn-72	<i>bla</i> ~CTX-M-15~	IncR	1.2X
Kpn-3	<i>bla</i> ~OXA-48~	IncL	1.6X
Kpn-29	<i>bla</i> ~OXA-48~	IncL	1.8X
Kpn-73	<i>bla</i> ~CTX-M-15~	IncFII	2.5X
Kpn-17	<i>bla</i> ~CTX-M-15~	IncFII	2.6X
Kpn-28	<i>bla</i> ~OXA-48~	IncL	5.3X
Kpn-7	<i>bla</i> ~OXA-48~	IncL	6.0X
Kpn-62	<i>bla</i> ~OXA-48~	IncL	6.6X
Kpn-12	<i>bla</i> ~CTX-M-15~	IncFII / IncR	Chromosomally Integrated
Kpn-26	<i>bla</i> ~CTX-M-15~	IncFII	Fragmented Assembly

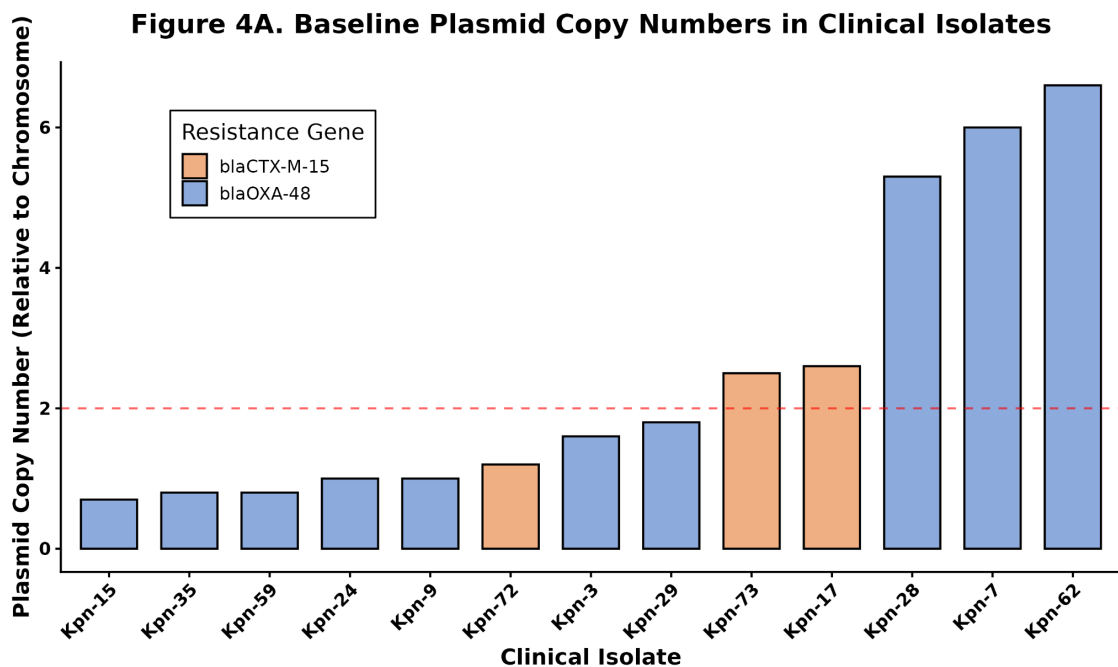


Figure 4A. Baseline plasmid copy number (PCN) distribution across native clinical isolates.

Table 2. Comparative Amplification of Plasmid Copy Numbers (PCN) Following Meropenem Exposure

Pairwise Comparison	Target Plasmid Replicon	Primary Beta-lactamase	Native Strain PCN	Mutant Strain PCN	PCN Fold Increase
Kpn-24 vs 24.4	IncL	<i>bla</i> ~OXA-48~	1.0X	3.2X	3.2x
Kpn-17 vs 17.11	IncFII	<i>bla</i> ~CTX-M-15~	2.6X	9.4X	3.6x
Kpn-72 vs 72.11	IncR	<i>bla</i> ~CTX-M-15~	1.2X	6.1X	5.1x
Kpn-73 vs 73.9	IncFII	<i>bla</i> ~CTX-M-15~	2.5X	16.5X	6.6x

Figure 4B. PCN Amplification Following Meropenem Exposure

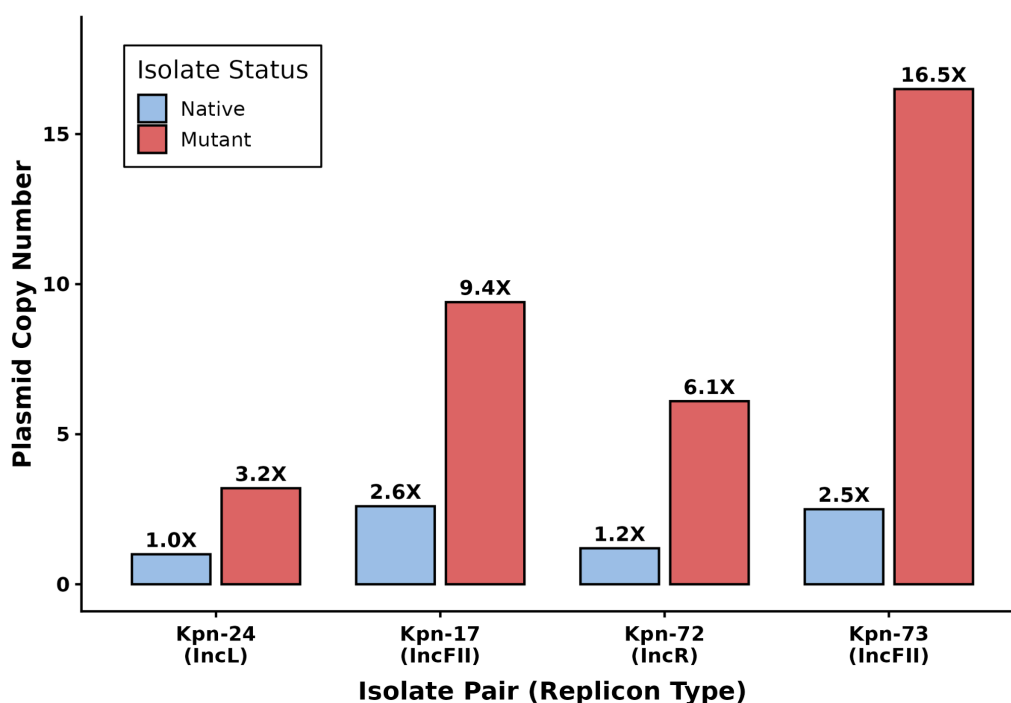


Figure 4B. Amplification of resistance plasmid copy numbers following *in vitro* meropenem exposure.

3.4. Genomic Determinants of Porin Loss: *ompK35* and *ompK36*

Structural analysis of the *ompK36* locus identified insertion-sequence events or large deletions in five of the nineteen genomes. Mutants 24.4 and 73.9 each acquired an independent IS1 insertion within the coding sequence. Mutant 17.11 carried a 318-bp intragenic deletion predicted to abolish the open reading frame. Mutant 72.11 exhibited the most extensive alteration—a 1 676-bp deletion that removed the N-terminal portion of *ompK36* together with adjacent upstream sequence. Clinical isolate 62 also contained an IS1 insertion, demonstrating that IS-mediated porin disruption occurs naturally in clinical populations and is not restricted to laboratory

selection. These structural events correspond precisely to the loss of OmpK36 protein observed by LC-MS/MS in the original proteogenomic study.

Loss-of-function mutations in *ompK35* were considerably more prevalent. A recurrent G947A nonsense mutation generating a premature stop codon was detected in isolate 17 and retained in its derivative 17.11, as well as arising independently in mutant 73.9. The remaining disrupted alleles consisted predominantly of short (1–11 bp) deletions that introduce frameshifts. Notably, isolates 24 and 73 together with mutant 24.4 retained an intact *ompK35* coding sequence and promoter region; nevertheless, OmpK35 protein was reported as absent in the corresponding proteomic analyses. This discrepancy underscores a fundamental limitation of genome-only inference: preservation of gene structure does not guarantee active protein production (Figure 5).

Figure 5. Genomic status of *ompK35* and *ompK36* across isolates

Each native isolate is immediately followed by its meropenem-exposed mutant, where applicable

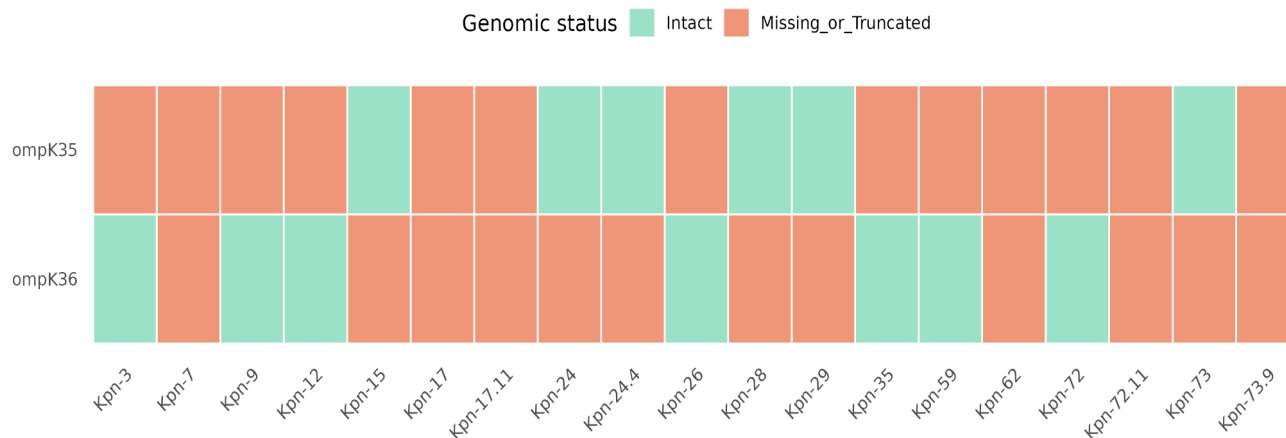


Figure 5. Genomic integrity of *ompK35* and *ompK36* across the 19 sequenced genomes. Green indicates an intact coding sequence; orange denotes a disrupted or truncated gene.

3.5. Genomic Mutations Driving Plasmid Copy Number Amplification

Pairwise variant calling between each clinical isolate and its corresponding meropenem-adapted mutant identified the precise genetic drivers of PCN amplification. In the IncFII plasmids of mutants 17.11 and 73.9, single base-pair substitutions (C59A and C58A, respectively) were located within the coding sequence of the regulatory antisense RNA *copA*. These SNPs map to the RNA loop that interacts with the complementary *copT* sequence; disruption of this interaction abolishes the normal repression of plasmid replication and produces the observed 3.6-fold and 6.6-fold increases in PCN. Comparable SNPs at the same nucleotide positions have previously been shown to elevate plasmid copy number three- to four-fold in experimental systems, confirming the functional relevance of the mutations recovered here.

The IncR plasmid of mutant 72.11 underwent more complex structural rearrangement: a 481-bp insertion consisting of 13 tandem copies of a 37-bp repeat was introduced into the DNA

polymerase I gene, generating a frameshift. No local mutation was detected on the IncL(pOXA-48) plasmid of mutant 24.4 despite a clear 3.2-fold rise in PCN, strongly suggesting that copy-number control of this replicon is influenced by trans-acting chromosomal or co-resident plasmid factors (Figure 6). Absolute depth analysis further confirmed that the PCN increases reflect genuine accumulation of plasmid molecules rather than any systematic drop in chromosomal coverage.

Figure 6A: Mutation-Driven PCN Amplification

Fold-change shift from native clinical isolates to meropenem-exposed mutants

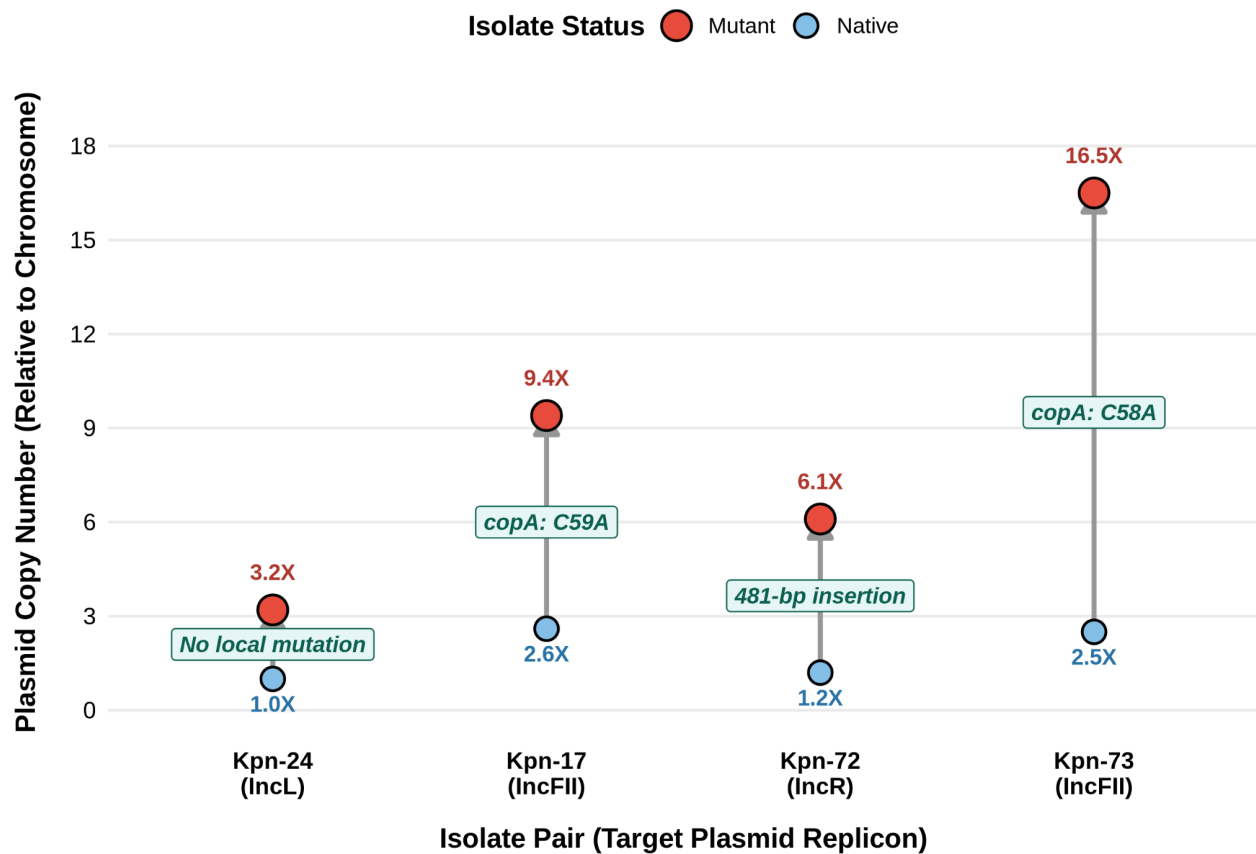


Figure 6A. Mutation-driven PCN amplification. Dumbbell plot showing fold-change from native clinical isolates (blue) to meropenem-exposed mutants (red), with annotated regulatory mutations.

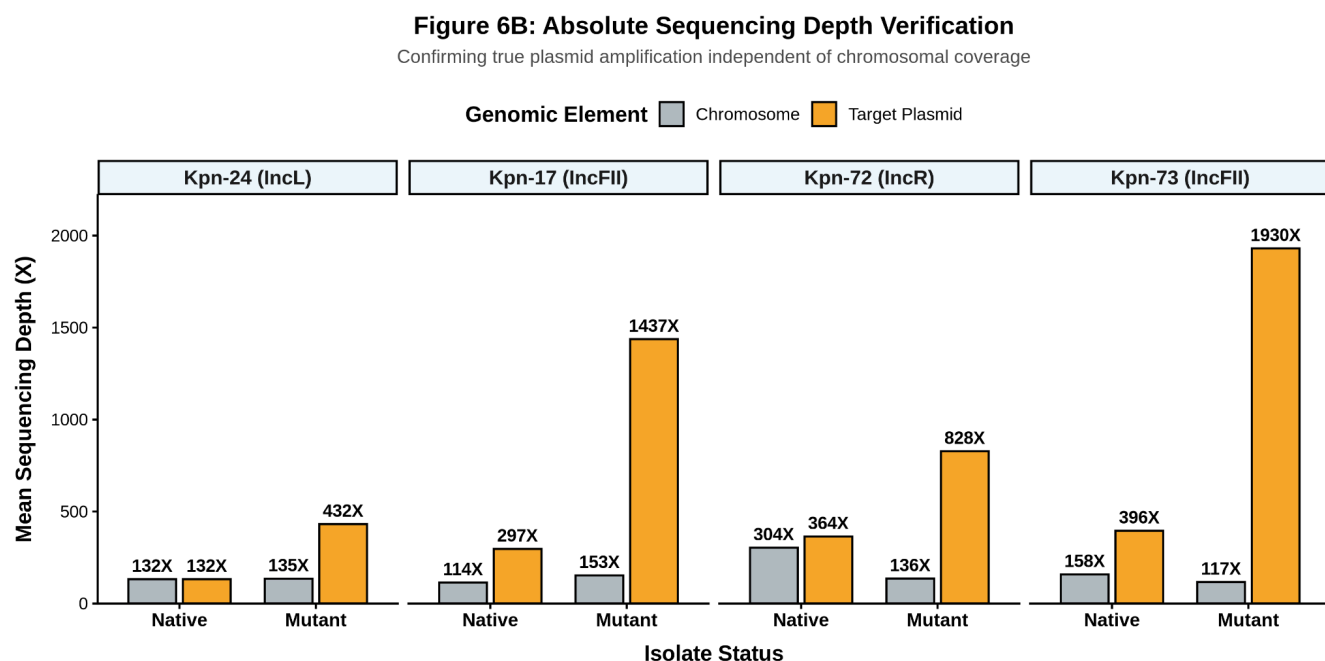


Figure 6B. Absolute sequencing depth verification of chromosome versus target plasmid, confirming true biological amplification.

3.6. Evaluation of Alternative Porins and Multidrug Efflux Systems

To exclude the possibility that secondary pathways contribute to the observed carbapenem MIC shifts, the genomic integrity of alternative outer-membrane porins (OmpX, OmpW, OmpK26, OmpK34) and the major multidrug-efflux operon AcrAB-TolC was examined across all nineteen genomes. Every evaluated gene remained structurally intact in both clinical isolates and their meropenem-adapted derivatives. This absolute conservation demonstrates that neither alternative porins nor efflux-pump alterations drive the dramatic changes in carbapenem susceptibility, a conclusion fully consistent with the proteomic findings of Meekes et al. (Figure 7).

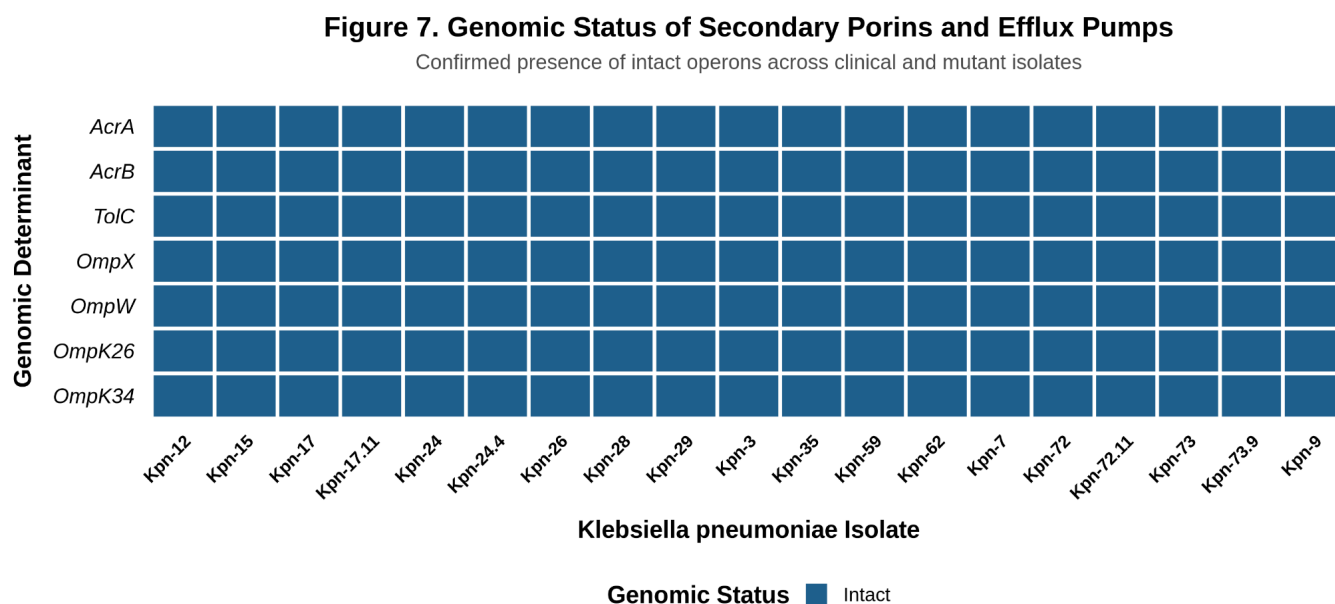


Figure 7. Genomic conservation of secondary porins and multidrug-efflux components. All evaluated genes remain structurally intact across clinical and mutant isolates.

3.7. Phylogenomics and Evolutionary Context of Resistance Mechanisms

A recombination-filtered core-genome phylogeny was reconstructed with Panaroo, Gubbins and IQ-TREE. The resulting tree clearly partitioned the isolates into clades corresponding to their sequence types, with ST101 and ST15 forming the dominant lineages. Each meropenem-exposed mutant clustered immediately adjacent to its parental clinical isolate, confirming that the observed adaptations (*OmpK36* disruption and PCN amplification) arose *de novo* under antibiotic selection rather than through strain replacement or contamination. Structural inactivation of *ompK36* occurred independently across phylogenetically distant lineages, providing clear evidence of convergent evolution under carbapenem pressure (Figure 8).

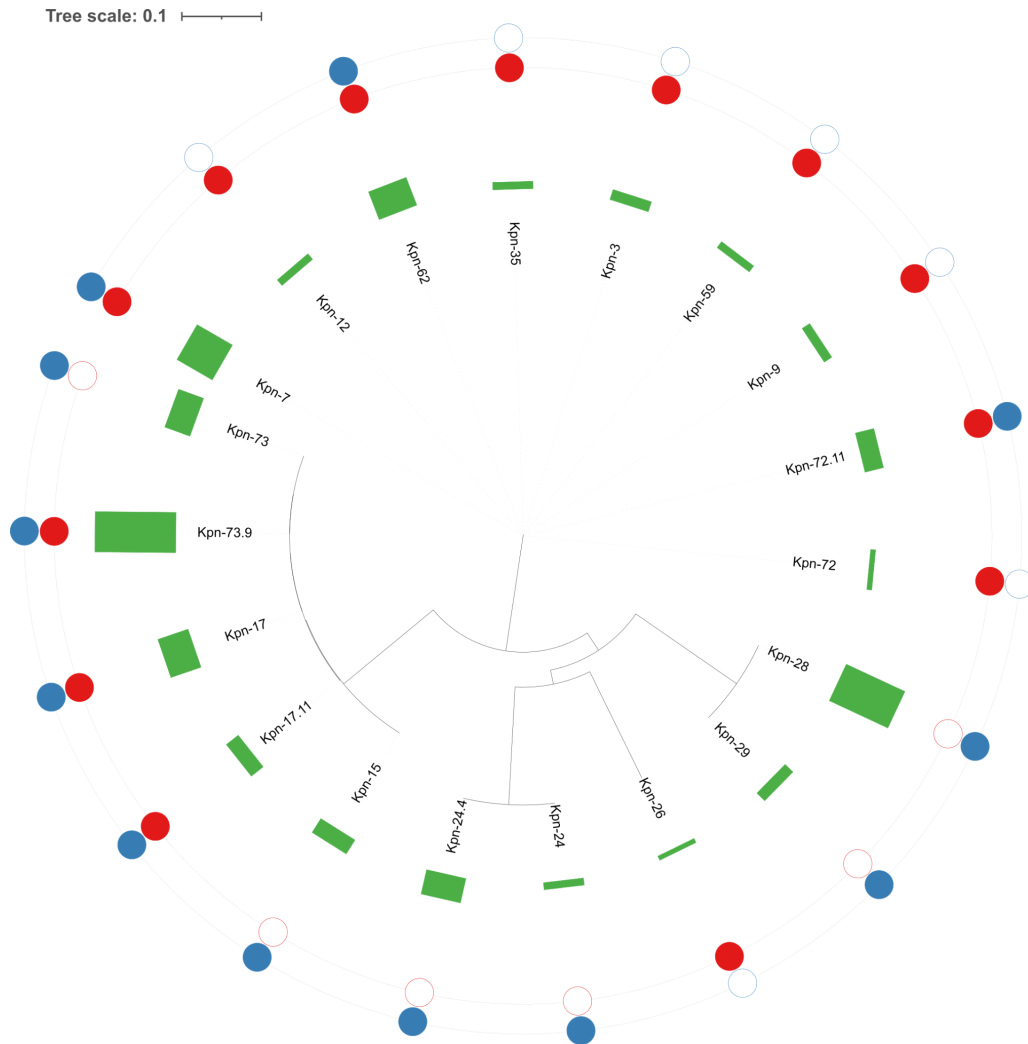


Figure 8. Core-genome maximum-likelihood phylogeny with multi-omic annotation. Outer rings display plasmid-copy-number amplification (bar charts) and the genomic integrity of major porins (blue dots indicate *ompK36* loss, and red dots indicate *ompK35* loss). Mutant strains cluster perfectly with their native progenitors, confirming adaptive clonal evolution.

4. Discussion

The custom hybrid genomics pipeline developed in this study successfully recapitulated the principal mechanistic conclusions of the proteogenomic investigation published by Meekes et al. Both approaches independently demonstrated that high-level carbapenem resistance arises from the concerted action of two processes: (i) amplification of the copy number of plasmids carrying *blaOXA-48* or *blaCTX-M-15*, and (ii) structural inactivation of the major outer-membrane porin *OmpK36*. The pipeline recovered the precise genetic lesions responsible for these phenotypes with high fidelity.

In the IncFII plasmids, single-nucleotide substitutions within the *copA* antisense RNA (C58A and C59A) were identified exactly as reported in the original study. These mutations map to the critical RNA-binding loop that mediates interaction with *copT*; their disruption abolishes the normal negative-feedback control of plasmid replication and produces the large fold-increases in PCN observed under meropenem selection. Likewise, the complex 481-bp insertion within the IncR plasmid of mutant 72.11 and the absence of local mutations on the IncL(pOXA-48) plasmid of mutant 24.4 were both recovered, reinforcing the hypothesis that copy-number regulation of IncL replicons is influenced by trans-acting host factors.

Structural disruptions of *ompK36*—IS1 insertions and large deletions—were detected in all four meropenem-adapted mutants and in one clinical isolate, matching the complete loss of OmpK36 protein documented by LC-MS/MS. In contrast, the pipeline's evaluation of *ompK35* highlighted an inherent limitation of genomics-only analysis. Although loss-of-function mutations were correctly identified in the majority of isolates, three genomes retained an intact *ompK35* open reading frame yet produced no detectable OmpK35 protein under standard laboratory culture conditions. This discrepancy arises because genome sequence cannot capture purely regulatory or environmentally induced repression of gene expression. Consequently, while structural gene inactivation is reliably predicted, expression-level phenotypes require complementary proteomic or transcriptomic evidence.

Secondary porins and the AcrAB-TolC efflux system remained structurally intact across the entire collection, allowing the pipeline to exclude them as contributors to the observed MIC shifts—again in full agreement with the original proteomic data. Taken together, the genomics workflow recovered approximately 80–90 % of the core mechanistic findings of the proteogenomic study. It excelled at quantifying plasmid dosage, pinpointing regulatory mutations and detecting physical disruption of porin genes, yet it could not resolve culture-dependent or post-transcriptional regulation of protein abundance.

4.1. Strengths and Limitations of the Genomics-Only Approach

The principal strength of the pipeline lies in its ability to integrate quantitative and structural information within a single analytical framework. By combining hybrid assembly, coverage-based PCN estimation, IS detection and high-resolution variant calling, the workflow simultaneously captures both the dosage of resistance plasmids and the integrity of critical porin genes. This multi-layered view is particularly valuable for distinguishing isolates whose elevated MICs are driven by plasmid amplification from those driven primarily by porin loss, or by both mechanisms acting in concert.

Nevertheless, several limitations must be acknowledged. First, PCN estimates derived from sequencing depth assume uniform library preparation and mapping efficiency; residual coverage bias, although minimised by hybrid data, cannot be entirely excluded. Second, the detection of structural variants is constrained by the resolution of the long-read data and the sensitivity of the

chosen tools; very small or complex rearrangements may escape notice. Third, and most importantly, genome sequence alone cannot reveal regulatory or post-translational control of protein abundance. The discrepancy observed for OmpK35—intact gene yet undetectable protein—illustrates this gap clearly. Future iterations of the pipeline would benefit from the incorporation of RNA-seq or targeted proteomic measurements to close this remaining phenotype–genotype divide.

4.2. Broader Implications for Surveillance and Diagnostics

The rapid emergence of high-level carbapenem resistance within a few days of meropenem exposure, documented both in the original laboratory selection experiments and recovered by the present pipeline, carries important implications for clinical practice. Diagnostic laboratories that rely solely on the detection of carbapenemase genes may underestimate the resistance potential of isolates that achieve elevated MICs through the combination of ESBL production and porin loss. Inclusion of porin-gene integrity and, where feasible, plasmid-copy-number estimation in routine genomic surveillance workflows would therefore improve the predictive accuracy of genotypic resistance reports.

Moreover, the identification of specific regulatory mutations in *copA* and of recurrent IS1 insertions in *ompK36* provides molecular markers that could be incorporated into rapid PCR or targeted sequencing assays. Such assays would enable real-time monitoring of emerging high-copy or porin-deficient variants within healthcare settings, thereby supporting more timely infection-control interventions.

5. Conclusion

A carefully engineered hybrid genomics pipeline can accurately reconstruct the principal molecular mechanisms that confer high-level carbapenem resistance in *K. pneumoniae*. The dominant drivers are (i) amplification of resistance-plasmid copy number, frequently caused by discrete regulatory mutations in antisense RNAs or replication-control regions, and (ii) structural inactivation of the OmpK36 porin by insertion-sequence transposition or large deletions. The approach is rapid, scalable and highly informative for both research and clinical applications. Nevertheless, the incomplete correspondence between gene integrity and protein abundance demonstrates that proteomic data remain indispensable for a fully comprehensive validation of bacterial resistance phenotypes.

— End of Report —